IUT 1 - 3

Exercises.

1. Assume $a^n - b^n \mid a^n + b^n$

Let M be min of number of 2 prime factors. For some $A, B \in \mathbb{N}^+$,

$$2^M(A^n - B^n) \mid 2^M(A^n + B^n)$$

$\Rightarrow A^n - B^n \mid A^n + B^n$

and at least one of A, B is odd.

From now on, assume one of a, b is odd. (*)

$$a^n + b^n \equiv 0 \mod a^n - b^n$$

$$-(a^n - b^n)$$

$$\equiv$$

$$2b^n$$

So, for some $k \in \mathbb{N}^+$, $2b^n = k(a^n - b^n)$

$$b^n = \frac{k}{k+2} a^n$$

$$\frac{b^n}{a^n} = \frac{k}{k+2}$$

So, for some $r \in \mathbb{N}^+$,

$$b^n = rk$$

$$a^n = r(k+2)$$

if one of a, b is even, other shauld be even.

This contradicts with (*).

So both a and b is odd.

Odd prime factors are only included.

For any odd prime p, $p \mid k \Rightarrow p \nmid k+2$

by checking $\pmod p$.

So, r is $\gcd(a^n, b^n)$. this is $\bigcirc^n$-formed.

Then, k and $k+2$ is also $\bigcirc^n$-formed number.

$n \geq 2$, and there're no such k.

$k \neq 1$ by easy check, then, if $k \geq 4$

if $k = u^n$ for some $u \in \mathbb{N}^+$ $(u \geq 2)$

$$
\begin{aligned}
k+2 &\geq (u+1)^n \\
&\geq u^n + nu^n \\
&\geq u^n + 2n \\
&\geq u^n + 4 \\
&= k+4
\end{aligned}
$$

binomial theorem

$n \geq 2$

※

Contradicts!

2. (a) By induction of k,

$k=1$ is obvious.

Assume $(a_1) + \cdots + (a_k) = (\gcd(a_1, \cdots, a_k))$

Let's prove $(a_1) + \cdots + (a_k) + (a_{k+1}) = (\gcd(a_1, \cdots, a_k, a_{k+1}))$

$$\begin{aligned} & \underline{(a_1) + \cdots + (a_k)} + (a_{k+1}) = (\underline{\gcd(a_1, \cdots, a_k, a_{k+1})}) \\ & \quad\parallel \\ & (\gcd(a_1, \cdots, a_k)) + (a_{k+1}) \\ & \quad\parallel \\ & (\underline{\gcd(\gcd(a_1, \cdots, a_k), a_{k+1})}) \end{aligned}$$

So it's enough to prove

$$\underbrace{\gcd(\underbrace{\gcd(a_1, \cdots, a_k)}_{= K}, a_{k+1})}_{= G} = \underbrace{\gcd(a_1, \cdots, a_k, a_{k+1})}_{= H}.$$

$$G \mid H) \quad \begin{cases} a_i \mid H \\ a_i \mid K \mid G \end{cases} \quad (i \in \{1, \cdots, k\})$$

$$\Rightarrow \quad K \mid H$$

$$a_{k+1} \mid H \qquad \text{so,} \qquad \underbrace{\gcd(K, a_{k+1})}_{= G.} \mid H$$

(Generally, $a \mid c$, $b \mid c \Rightarrow \gcd(a, b) \mid c$)

$H \mid G)$ $\begin{cases} a_i \mid K \mid G & (i \in \{1, \dots, k\}) \\ a_{k+1} \mid G \end{cases}$

$$\Rightarrow \quad a_i \mid G \quad (i \in \{1, \dots, k\}).$$

So, $H \mid G$.

Thus, $\underline{G = H}$ $\blacksquare$

$$\left(\text{Generally, } a_1 \mid c, \dots, a_n \mid c \Rightarrow \gcd(a_1, \dots, a_n) \mid c \right).$$

(b) Stmt.

$\exists x_1, \cdots, x_n$

$$a_1 x_1 + \cdots + a_n x_n = c \iff \gcd(a_1, \cdots, a_n) \mid c$$

Prf

($\Rightarrow$) By induction of n, $n=1$ is obvious.

Assume

$\exists x_1, \cdots x_n, x_{n+1}$. $\underline{a_1 x_1 + \cdots + a_n x_n + a_{n+1} x_{n+1} = c}$

$\iff a_1 x_1 + \cdots + a_n x_n = c - a_{n+1} x_{n+1}$.

By Induction premise, $\gcd(a_1, \cdots, a_n) \mid c - a_{n+1} x_{n+1}$

$$\gcd(a_1, \cdots, a_n, a_{n+1}) \mid \gcd(a_1, \cdots, a_n) \mid c - a_{n+1}x_{n+1}$$

and $$\gcd(a_1, \cdots, a_n, a_{n+1}) \mid a_{n+1}$$

So $$\gcd(a_1, \cdots, a_n, a_{n+1}) \mid c$$

$\left(\text{By checking } \left(\text{mod } \gcd(a_1, \cdots, a_n, a_{n+1})\right)\right)$ //

($\Leftarrow$) By induction of n.

Assume $\underbrace{\gcd(a_1, \cdots, a_n, a_{n+1})}_{\gcd(\gcd(a_1, \cdots, a_n), a_{n+1})} \mid c$.

By induction premise for $n=2$, or just Euclidian method,

$\exists\, y, x_{n+1}$. $y \cdot \gcd(a_1, \cdots, a_n) + x_{n+1} a_{n+1} = c$

B induction premise again, $\exists\, z_1, \cdots, z_n$,

$$z_1 a_1 + \cdots + z_n a_n = \gcd(a_1, \cdots, a_n)$$

Define $x_i := y z_i$ $(i \in \{1, \cdots n\})$,

We got $\underline{x_1 a_1 + \cdots + x_n a_n + x_{n+1} a_{n+1} = C}$

3. (a) Assume $\deg g \neq \deg f$.

$g = fq$ for some $q \in A$.

$\deg g = \deg f + \deg q$.

$\deg q \neq 0$, so $\deg q \geq 2$

Thus we can get $2 + \deg f \leq \deg g$.

(b) $f \mid x^5, x^6$

By $f \mid x^5$, $\deg f \leq 5$.

if $\deg f = 4$, $f \nmid x^5$ by (a)

if $\deg f = 5$, $f \nmid x^6$ by (a)

Thus $\deg f \leq 3$ ∎

(C) Assume f exists as

$$x^2, x^3 \mid f \mid x^5, x^6$$

By $x^3 \mid f$ and (2), $3 \leq \deg f \leq 3$ so $\deg f = 3$.

But $x^2 \mid f$, and this contradicts with (a) ∎

4. (a) $(0) = \{0\}$,

if $ab=0$, $\mathbb{Z}$ is integral domain so $a=0$ or $b=0$

So, (0) is prime.

$(0) \subsetneq (2) \neq \mathbb{Z}$, so it's not maximal ■

(b) $(x) = \mathbb{Z}[x] \setminus \{a \mid a \in \mathbb{Z}, a \neq 0\} = \{a \mid a \in \mathbb{Z}[x], \deg a \neq 1\}$

if $ab \in (x)$, if $ab=0$, same as (a).

if $ab \neq 0$, $\deg ab \geq 1$, so $\deg a \geq 1$ or $\deg b \geq 1$

This means $a \in (x)$ or $b \in (x)$

Thus (x) is prime.

$(x) \subsetneq (2, x) \neq \mathbb{Z}[x]$ so it's not maximal. $\blacksquare$